

STRUCTURE PROPERTIES

FORMULAS FOR GEOMETRICAL PROPERTIES OF SECTIONS

Cross-section area

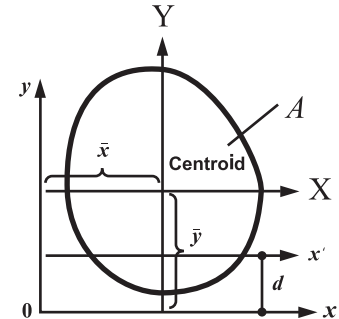
$$A = \int_A dA$$

First moment

$$S_x = \int_A y dA, S_y = \int_A x dA$$

Position of centroid

$$\bar{x} = \frac{S_y}{A}, \bar{y} = \frac{S_x}{A}$$



Moment of inertia

$$J_x = \int_A y^2 dA, J_y = \int_A x^2 dA$$

Product of inertia

$$J_{xy} = \int_A xy dA$$

Polar moment of inertia

$$J_p = \int_A r^2 dA = J_x + J_y$$

Radius of gyration

$$i_x = \sqrt{\frac{J_x}{A}}, i_y = \sqrt{\frac{J_y}{A}}$$

Polar radius of gyration

$$i_p = \sqrt{\frac{J_p}{A}}$$

Section modulus

$$Z_1 = \frac{J}{y_1}, Z_2 = \frac{J}{y_2}$$

Transition of axis

$$S_x' = S_x - Ad, S_y' = Ay - Ad$$

$$J_x' = J_x + Ay^2, J_y' = J_y + Ax^2, J_{xy}' = J_{xy} + xAy$$

Rotation of axis

$$S_x' = S_x \cos \alpha - S_y \sin \alpha$$

$$J_x' = J_x \sin^2 \alpha + J_y \cos^2 \alpha + J_{xy} \sin 2\alpha$$

$$J_x' + J_y' = J_x + J_y = J_p$$

